MONO65

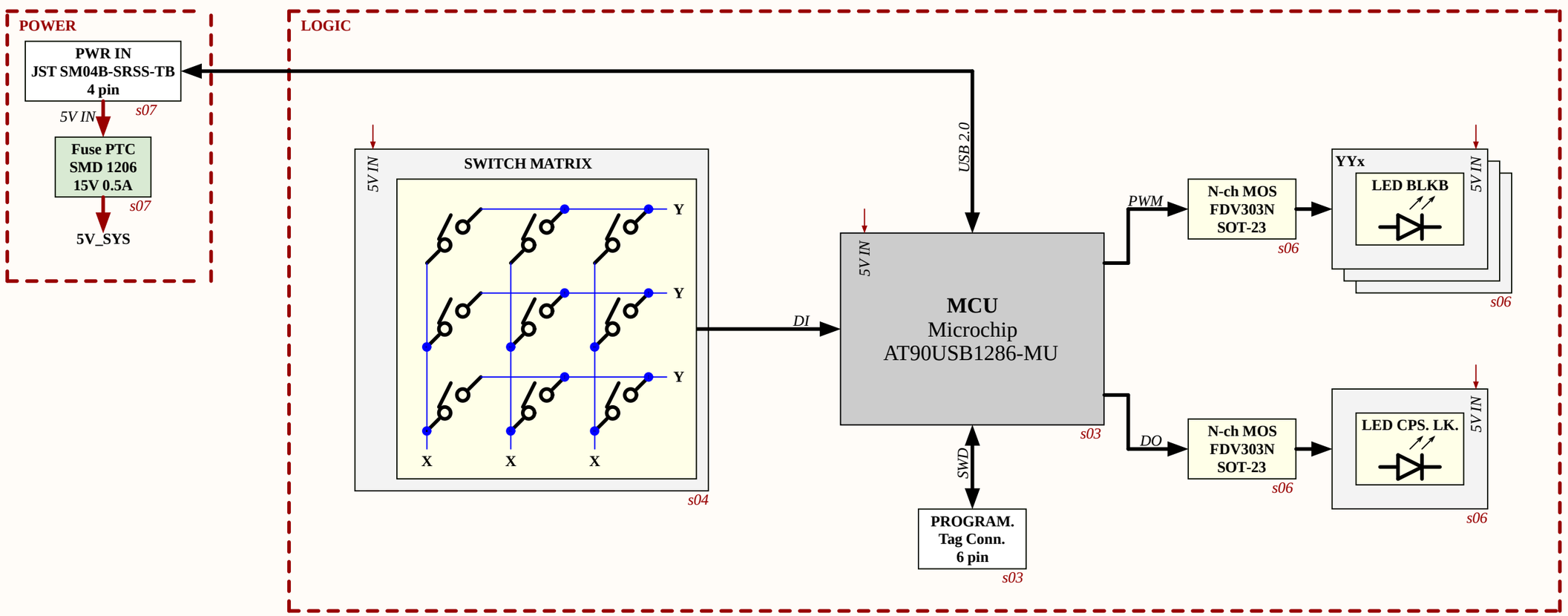
Version: A2

Assembly: NO_SWITCHES

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Power line load
Power line source
Data line

Connectivity
Peripheral
Main core
Power supply

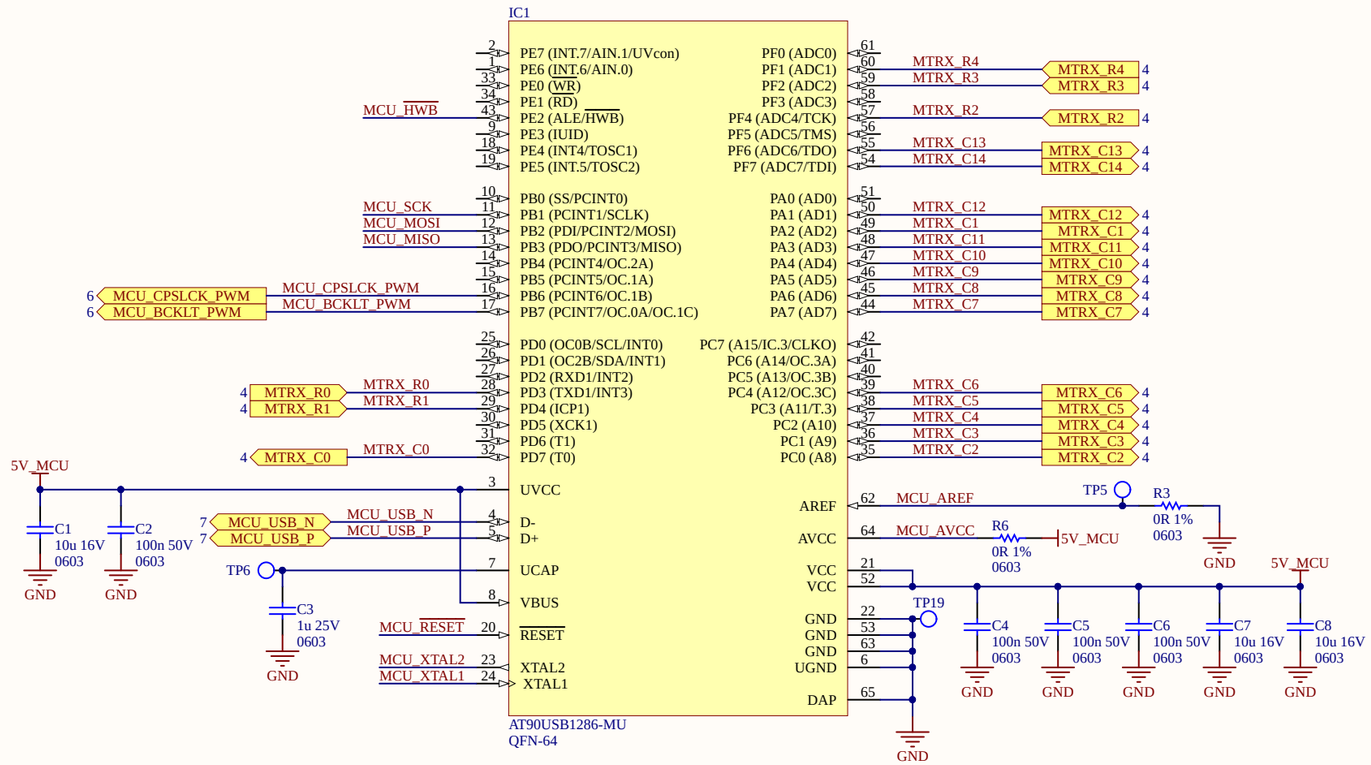
CLOSED SOURCE

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Page Contents: BLOCK DIAGRAM		
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MCU AT90USB1286

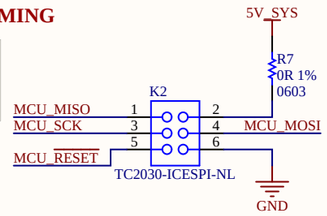
MCU 8-bit AVR, 16MHz, 128KB Flash, 64-QFN 9x9mm (PS package).

ISP & USB programming.



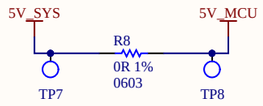
ISP (SPI) PROGRAMMING

The 10-pin male connector on Atmel-ICE has reversed pins. For Atmel-ICE use the LEMTA version of Tag Connect cables.



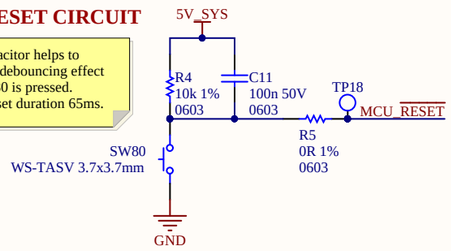
MCU POWER JUMPER

0R 0603 jumper resistor is used for debugging to disconnect the MCU from the system.



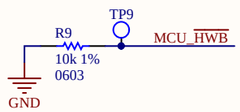
MCU RESET CIRCUIT

100nF capacitor helps to reduce the debouncing effect while SW80 is pressed.
Internal reset duration 65ms.



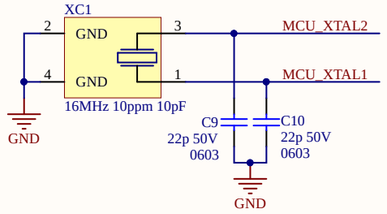
BOOTLOADER ACTIVATION

HWB allows to execute the bootloader after reset when tied to GND during external reset pulse.
The HWB mode of this pin is active only when the HWBE fuse is enable.



MCU CRYSTAL OSCILLATOR

Place capacitors close to the crystal oscillator. Connect the oscillator and capacitors to the same GND polygon.



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MX SWITCHES

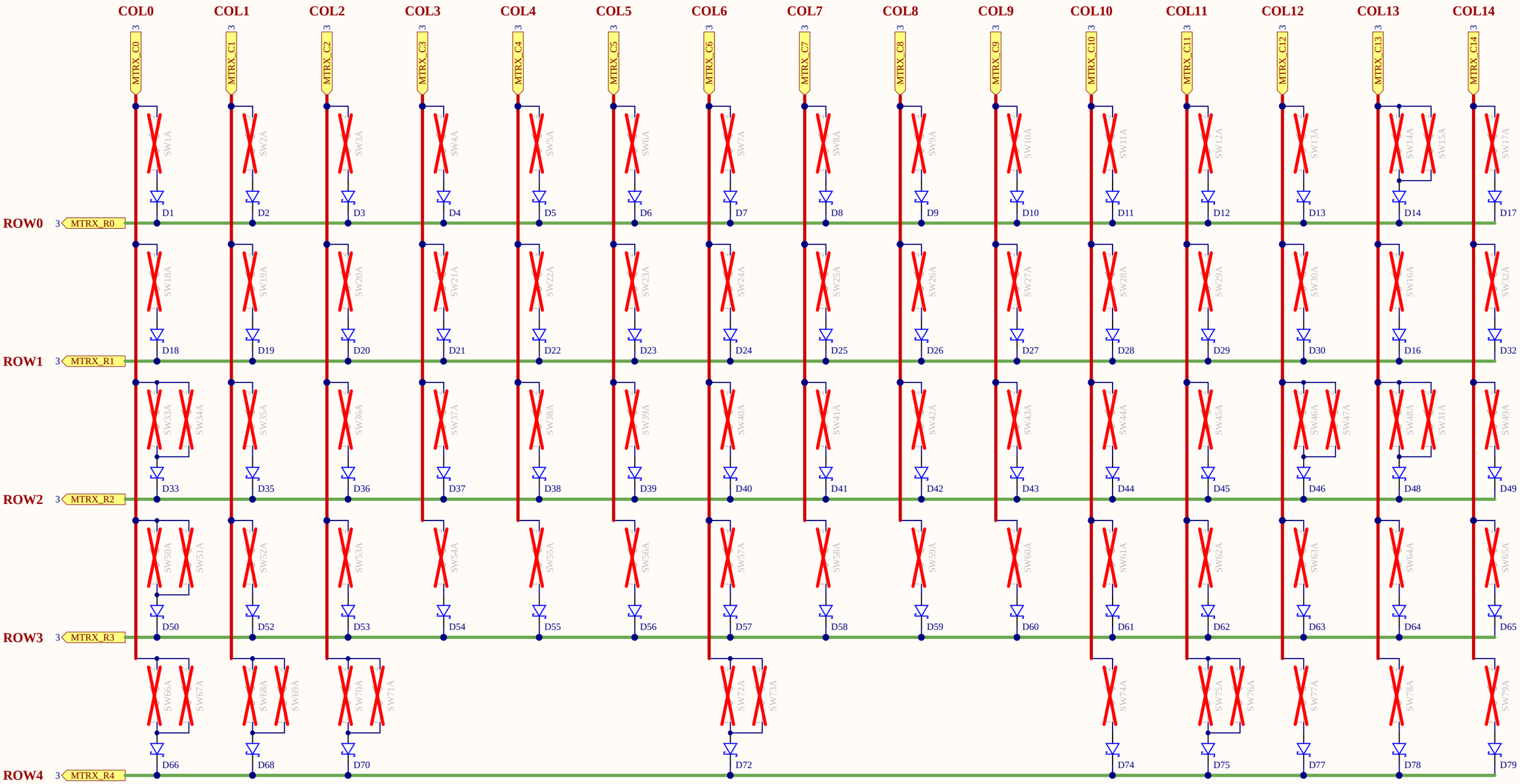
Matrix of MX Switches.

Switches supports backlight LEDs - located on the "s06 [LEDs]".

Switches with the footprint “MX1A-1-reversed LED-Pins”: SW67, SW69, SW71 and SW76.

Switches with the footprint “MX1A-1-LED-Pins”: all others.

For switch pairs SW14/SW15, SW46/SW47, SW48/SW31, SW50/SW51, SW66/SW67, SW68/SW69, SW70/SW71, SW72/SW73, SW75/SW76 only one switch/LED is used -> same resistor and same diode.



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BACKLIGHT LEDS

Backlight LEDs.

Backlight LEDs are PWM controlled.

For switch pairs SW14/SW15, SW46/SW47, SW48/SW31, SW50/SW51, SW66/SW67, SW68/SW69, SW70/SW71, SW72/SW73, SW75/SW76 only one switch/LED is used -> same resistor and same diode.

Resistor calculated for Nichia NSPW310DS LED.
Vfwd = 2.9V@5mA



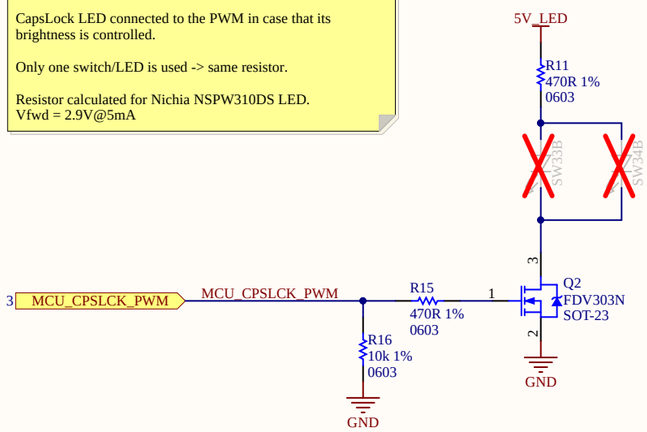
CAPS LOCK LED

CapsLock LED.

CapsLock LED connected to the PWM in case that its brightness is controlled.

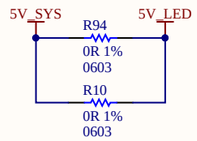
Only one switch/LED is used -> same resistor.

Resistor calculated for Nichia NSPW310DS LED.
Vfwd = 2.9V@5mA



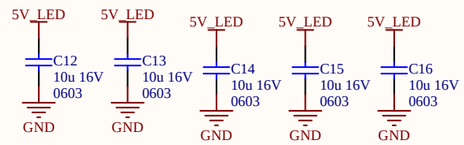
LED POWER JUMPER

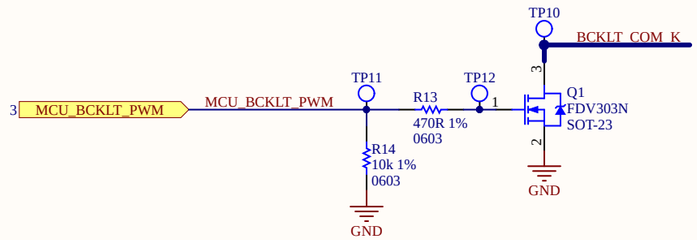
0R 0603 jumper resistor is used for debugging to disconnect LEDs from the system.



LED POWER BUFFER

Place one 10uF capacitor per LED line.





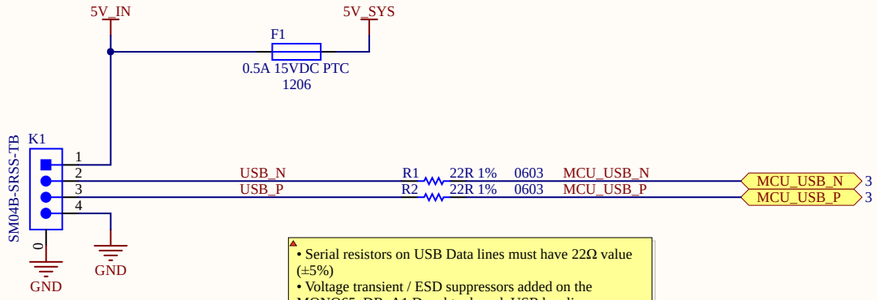
CLOSED SOURCE

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INPUT CONNECTOR

Input connector JST deliveres 5V from the MONO65_DB_A1 Daughter board.

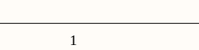
5V_SYS protected with a PTC fuse.
500mA holding current.
1.0A trip current.



• Serial resistors on USB Data lines must have 22Ω value (±5%)
• Voltage transient / ESD suppressors added on the MONO65_DB_A1 Daughter board. USB bus line

PNP FIDUCIALS

Place PnP fiducials on the same side as SMD parts.
Two of them close to MCU.



PWR TEST POINTS

Place test points on the same side as SMD parts.

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